Distribution of a random variable

A random variable for a chance experiment is a function $X : \Omega \to R$ whose inputs are outcomes of the experiment and whose outputs are real numbers.

The distribution of a random variable is a list of its possible values and, for each possible value x, the probability P(X = x) of the event “X = x”.

Describe the distributions of the following random variables: list the possible values of the random variable and calculate the probability that each of these values occurs.

Roll a single die. The random variable X is the number rolled.

Possible values 1, 2, 3, 4, 5, 6.

P(X = 1) = 1/6, P(X = 2) = 1/6, P(X = 3) = 1/6, P(X = 4) = 1/6, P(X = 5) = 1/6, P(X = 6) = 1/6.

Roll two dice (red and green). The random variable X is the higher of the two numbers rolled.

Roll two dice. The random variable X is the sum of the numbers rolled.

Flip a coin five times. The random variable N is the number of heads that you get.

Draw a poker hand (five cards from a standard deck of 52 cards). The random variable N is the number of aces in your hand.